

Free Energy & Kc 1

Essential Pre-Uni Chemistry L2.1

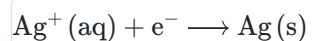
Subject & topics: Chemistry | Physical | Electrochemistry **Stage & difficulty:** A Level P2

Reduction	$E^\ominus / \text{V}$
$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \longrightarrow \text{Zn}(\text{s})$	-0.76
$\text{Cr}^{3+}(\text{aq}) + 3\text{e}^- \longrightarrow \text{Cr}(\text{s})$	-0.74
$\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \longrightarrow \text{Fe}(\text{s})$	-0.44
$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \longrightarrow \text{Cu}(\text{s})$	+0.34
$\text{Cu}^+(\text{aq}) + \text{e}^- \longrightarrow \text{Cu}(\text{s})$	+0.52
$\text{Fe}^{3+}(\text{aq}) + \text{e}^- \longrightarrow \text{Fe}^{2+}(\text{aq})$	+0.77
$\text{Ag}^+(\text{aq}) + \text{e}^- \longrightarrow \text{Ag}(\text{s})$	+0.80
$\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 6\text{e}^- + 14\text{H}^+(\text{aq}) \longrightarrow 2\text{Cr}^{3+}(\text{aq}) + 7\text{H}_2\text{O}(\text{l})$	+1.33

Use the standard electrode potentials tabulated above to find $\Delta G^\ominus$ for the following reactions:

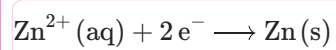
Part A

(a)



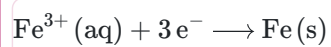
Part B

(b)



Part C

(c)



Question deck:

STEM SMART Chemistry Week 45

Free Energy & Kc 2

Essential Pre-Uni Chemistry L2.2

Subject & topics: Chemistry | Physical | Electrochemistry **Stage & difficulty:** A Level P2

Reduction	$E^\ominus / \text{V}$
$\text{Zn}^{2+}(\text{aq}) + 2\text{e}^- \longrightarrow \text{Zn}(\text{s})$	-0.76
$\text{Cr}^{3+}(\text{aq}) + 3\text{e}^- \longrightarrow \text{Cr}(\text{s})$	-0.74
$\text{Fe}^{2+}(\text{aq}) + 2\text{e}^- \longrightarrow \text{Fe}(\text{s})$	-0.44
$\text{Cu}^{2+}(\text{aq}) + 2\text{e}^- \longrightarrow \text{Cu}(\text{s})$	+0.34
$\text{Cu}^+(\text{aq}) + \text{e}^- \longrightarrow \text{Cu}(\text{s})$	+0.52
$\text{Fe}^{3+}(\text{aq}) + \text{e}^- \longrightarrow \text{Fe}^{2+}(\text{aq})$	+0.77
$\text{Ag}^+(\text{aq}) + \text{e}^- \longrightarrow \text{Ag}(\text{s})$	+0.80
$\text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 6\text{e}^- + 14\text{H}^+(\text{aq}) \longrightarrow 2\text{Cr}^{3+}(\text{aq}) + 7\text{H}_2\text{O}(\text{l})$	+1.33

Use the standard electrode potentials tabulated above to find $\Delta G^\ominus$ for the following reactions:

Part A

(a)

$\text{Ag}^+(\text{aq}) + \text{Fe}^{2+}(\text{aq}) \longrightarrow \text{Fe}^{3+}(\text{aq}) + \text{Ag}(\text{s})$. Give your answer to 1 significant figure.

Part B

(b)

$3 \text{Zn(s)} + \text{Cr}_2\text{O}_7^{2-}(\text{aq}) + 14 \text{H}^+(\text{aq}) \longrightarrow 3 \text{Zn}^{2+}(\text{aq}) + 2 \text{Cr}^{3+}(\text{aq}) + 7 \text{H}_2\text{O(l)}$. Give your answer to 3 significant figures.

Part C

(c)

$2 \text{Cr(s)} + 3 \text{Cu}^{2+}(\text{aq}) \longrightarrow 2 \text{Cr}^{3+}(\text{aq}) + 3 \text{Cu(s)}$. Give your answer to 3 significant figures.

Question deck:

STEM SMART Chemistry Week 45

Buffers 4

Essential Pre-Uni Chemistry J5.4

Subject & topics: Chemistry | Physical | Acids & Bases **Stage & difficulty:** A Level P2

Part A

(a)

Given that methanoic acid has a K_a of $1.6 \times 10^{-4} \text{ mol dm}^{-3}$, calculate the pH of a solution containing 25 mmol of methanoic acid and 40 mmol of potassium methanoate.

Part B

(b)

Given that methanoic acid has a K_a of $1.6 \times 10^{-4} \text{ mol dm}^{-3}$, calculate the pH of a solution containing 0.40 mol of methanoic acid and 0.32 mol of magnesium methanoate.

Question deck:

STEM SMART Chemistry Week 45

Buffers 7

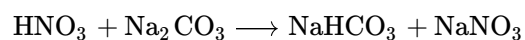
Essential Pre-Uni Chemistry J5.7

Subject & topics: Chemistry | Physical | Acids & Bases **Stage & difficulty:** A Level P3

N.B. This question is significantly harder

Calculate the mass of sodium carbonate decahydrate, $\text{Na}_2\text{CO}_3 \cdot 10 \text{H}_2\text{O}$, that should be added to 2.5 dm^3 of 0.40 mol dm^{-3} nitric acid to make a buffer of pH 10.5, given that the $\text{p}K_{\text{a}}$ of hydrogencarbonate is 10.3.

Assume: (1) That no CO_2 is given off in the reaction. (2) That the nitric acid just determines the hydrogencarbonate concentration and does not participate in the buffer, so that the hydrogencarbonate concentration is obtained from the reaction,



Question deck:

STEM SMART Chemistry Week 45

Buffers 9

Essential Pre-Uni Chemistry J5.9

Subject & topics: Chemistry | Physical | Acids & Bases **Stage & difficulty:** A Level P3

A buffer made from "hexamine" containing $0.00250 \text{ mol dm}^{-3}$ of hexamine and $0.00180 \text{ mol dm}^{-3}$ of its conjugate acid has a pH of 5.03. Calculate the pK_b of hexamine.

Question deck:

STEM SMART Chemistry Week 45

Buffers 10

Essential Pre-Uni Chemistry J5.10

Subject & topics: Chemistry | Physical | Acids & Bases **Stage & difficulty:** A Level P3

A buffer of pH 7.8 is prepared by taking 200 cm^3 of 0.020 mol dm^{-3} "tris" solution and adding dilute hydrochloric acid from a burette until the pH is correct. If this requires 1.35 cm^3 of 2.0 mol dm^{-3} HCl(aq) , calculate the $\text{p}K_{\text{a}}$ of the conjugate acid of "tris".

Question deck:

STEM SMART Chemistry Week 45

2-Hydroxybenzoate Complex

Subject & topics: Chemistry | Inorganic | Transition Metals **Stage & difficulty:** A Level C1

Mixtures of solutions containing Fe^{3+} (aq) and 2-hydroxybenzoate ions show light absorbancy as shown in the graph below.

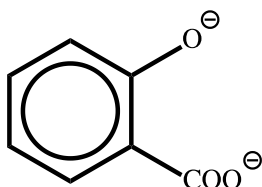


Figure 1: 2-Hydroxybenzoate ion

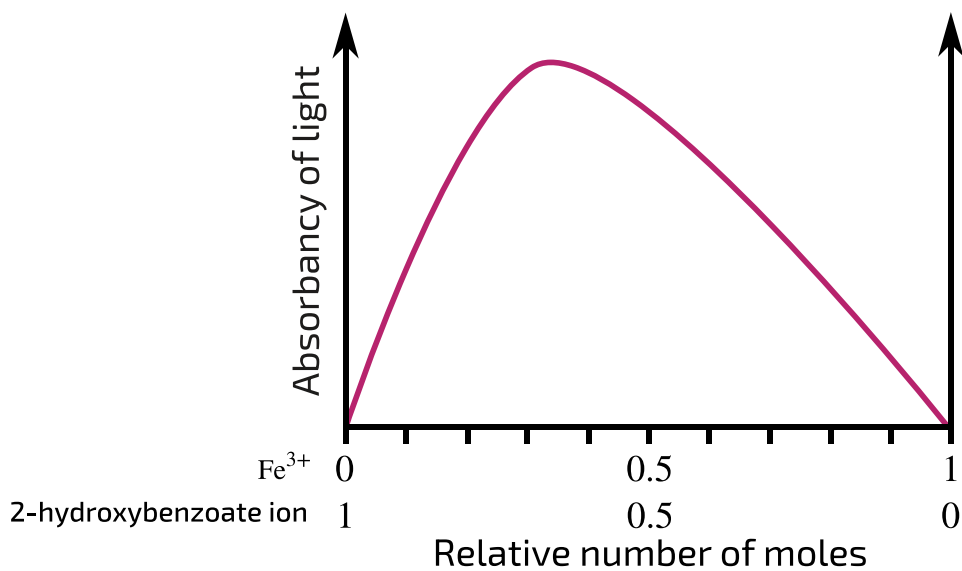


Figure 2: Variation of light absorbance with relative moles of Fe^{3+} and 2-hydroxybenzoate.

Part A

Ratio

What is the ratio of these two species (Fe^{3+} : 2-hydroxybenzoate) in the complex ion?

- ☐ 1:2
- ☐ 1:1
- ☐ 1:4
- ☐ 3:1
- ☐ 1:3
- ☐ 2:1
- ☐ 4:1

Part B

Coordination number

Suggest what is the likely coordination number of iron in this ion.

Part C

Coordination geometry

Suggest a coordination geometry for the Fe^{3+} -2-hydroxybenzoate complex.

Part D
Charge

What will be the overall charge on the complex ion?

Adapted with permission from UCLES A-Level Chemistry 1988, Paper 2, Question 7.

Question deck:

STEM SMART Chemistry Week 45

Arrhenius Equation

Subject & topics: Chemistry | Physical | Kinetics **Stage & difficulty:** A Level C2

The temperature of a sample of hydrogen iodide is raised from 300°C to 500°C . The activation energy for the dissociation of hydrogen iodide ($2\text{HI} \longrightarrow \text{H}_2 + \text{I}_2$) is 190 kJ mol^{-1} .

Part A

Rate constants

Calculate the ratio of the rate constants at the two temperatures (as a number greater than 1, to 2 significant figures).

Part B

Average velocity

Calculate the ratio of the average velocities of the molecules at the two temperatures (to 2 significant figures).

Part C**Activation energy**

The enzyme glucose-6-phosphate dehydrogenase catalyses the reaction:

Given the data below and the Arrhenius equation $k = Ae^{-\frac{E_a}{RT}}$, use a graphical method to calculate the activation energy of the reaction, E_a .

$T / ^\circ\text{C}$	$k / \text{M}^{-1} \text{s}^{-1}$
30	1.95
35	2.40
40	2.82
45	3.31
50	3.89

Adapted with permission from UCLES, A Level Chemistry, November 1968, Paper 2, Question 2. Last part used with permission from the Cambridge Chemistry Challenge : C3L6

Question deck:

STEM SMART Chemistry Week 45

Aromatic Acids and Bases

Subject & topics: Chemistry | Organic | Aromaticity **Stage & difficulty:** A Level C2

The benzene ring has a strong effect on the acidity and basicity of groups directly bonded to the ring. **Negative charges** and **lone pairs** of electrons are **delocalised** into the aromatic ring.

OH groups become more acidic and N lone pairs become less basic as a result of this delocalisation.

Part A

Aromatic acids

Match the aromatic compound with its correct pK_a value.

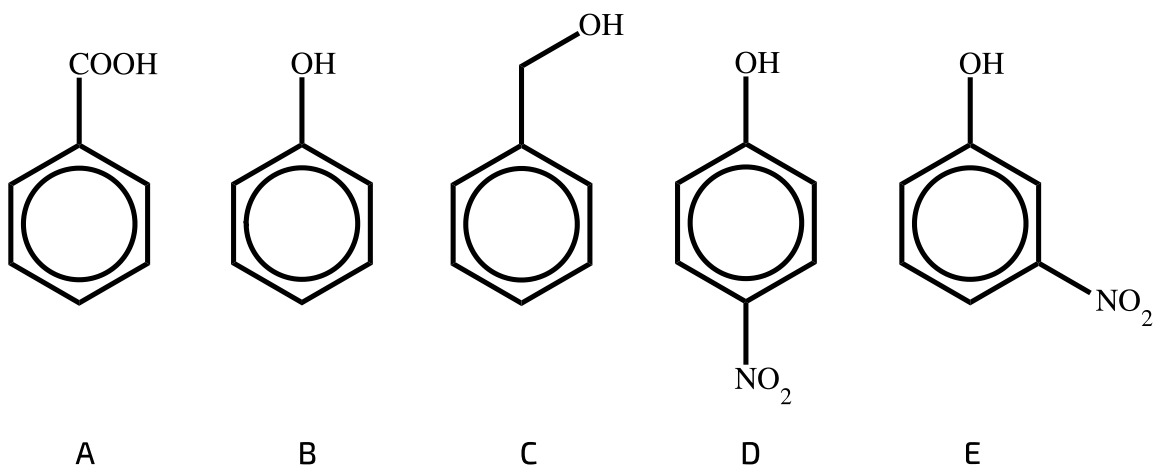


Figure 1: Aromatic compounds A - E with acidic hydrogen atoms.

Compound	pK_a
A	
B	
C	
D	
E	

Items:

Part B

Phenylamine with aqueous sulfuric acid

Which of the following products is formed when phenylamine (aniline) dissolves in dilute aqueous sulfuric acid?

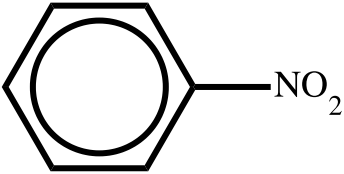
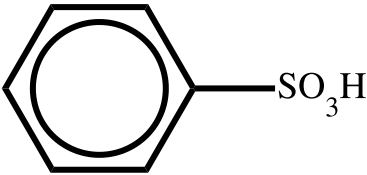
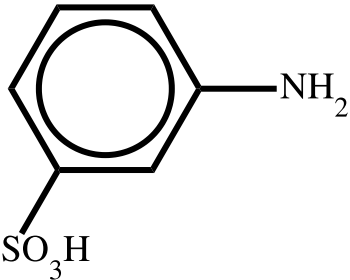
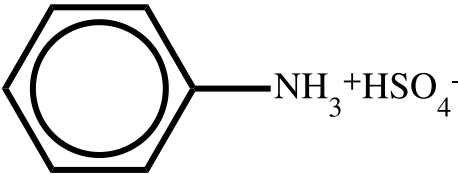
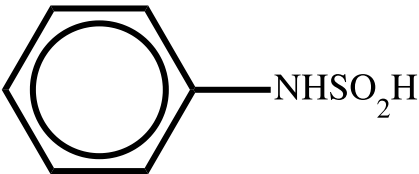
- A 
- B 
- C 
- D 
- E 

Figure 2: Possible products of phenylamine with dilute aqueous sulfuric acid.

- ☐ A
- ☐ B
- ☐ C
- ☐ D
- ☐ E

Part C

Basicity of amines

Rank the following compounds in order of increasing basicity:

Most basic

Least basic

Items:

Ethylamine

Diphenylamine

Phenylamine

Ammonia

Diethylamine

Parts A and C created for isaacphysics.org by R. Less;

Part B adapted with permission from OCSEB, A-Level Chemistry, June 1994, Paper 1, Question 28

Question deck:

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